



NeuroSpace

Complete evaluation export

Job ID	577c7f4c-f11c-4b11-af66-ff333b127996
Status	completed
Result timestamp	2026-05-15T20:24:47.115239
PDF generated	2026-05-15 21:30:09

Pipeline Configuration

Parameter	Value
model_types	rnn, qrnn
comparison_mode	pca
feature_selector	pca
manual_overrides.rnn_epochs	20
manual_overrides.qrnn_epochs	10
manual_overrides.batch_size	32
manual_overrides.rnn_batch_size	32
manual_overrides.qrnn_batch_size	16
num_classes	2
class_names	0, 1
qrnn_backend	qiskit
noise_enabled	False
noise_level	0.0000
mitigation_enabled	False
mitigation_runs	1

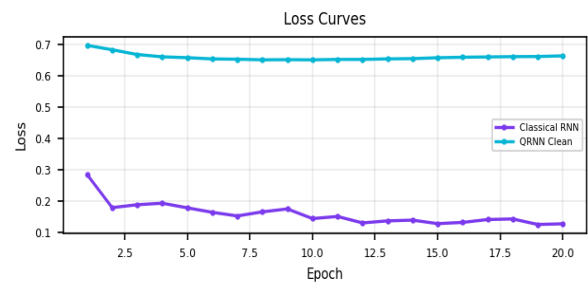
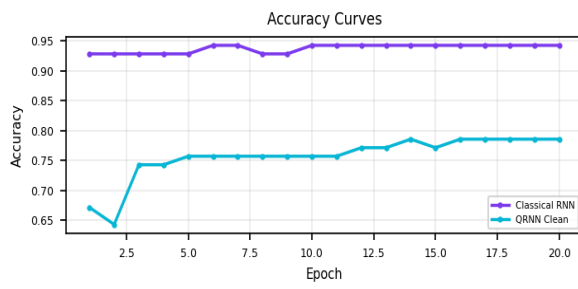
Best Model

Winner	RNN Accuracy	QRNN Accuracy	Improvement
RNN	96.43%	65.71%	30.71%

Evaluation Summary

Model	Accuracy	Precision	Recall	F1 Score	Loss
Classical RNN	96.43%	96.57%	96.43%	96.45%	0.1277
QRNN Clean	65.71%	68.53%	65.71%	66.48%	0.6643

Learning Curves



Classical RNN

Metric	Value
Accuracy	96.43%
Precision	96.57%
Recall	96.43%
F1 Score	96.45%
Loss	0.1277
Train Loss	0.0620
Val Loss	0.1277
Train Accuracy	98.57%
Val Accuracy	94.29%
Train Size	489
Val Size	70
Test Size	140

Classical RNN Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	88	4
Class 1	1	47

Classical RNN Classification Report

Class	Precision	Recall	F1 Score	Support
0	98.88%	95.65%	97.24%	92.0000
1	92.16%	97.92%	94.95%	48.0000
macro avg	95.52%	96.78%	96.09%	140.0000
weighted avg	96.57%	96.43%	96.45%	140.0000
Accuracy			96.43%	

Classical RNN Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.3266	0.2849	87.53%	92.86%
2	0.1391	0.1794	95.71%	92.86%
3	0.1319	0.1887	96.11%	92.86%
4	0.0946	0.1935	98.16%	92.86%
5	0.1042	0.1788	97.34%	92.86%
6	0.0814	0.1644	97.55%	94.29%
7	0.0964	0.1529	97.96%	94.29%
8	0.0856	0.1660	97.55%	92.86%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
9	0.0822	0.1752	97.96%	92.86%
10	0.0706	0.1445	97.96%	94.29%
11	0.0848	0.1511	97.55%	94.29%
12	0.0682	0.1307	97.75%	94.29%
13	0.0841	0.1371	97.96%	94.29%
14	0.0650	0.1395	97.96%	94.29%
15	0.0721	0.1283	97.34%	94.29%
16	0.0586	0.1322	98.36%	94.29%
17	0.0627	0.1414	98.36%	94.29%
18	0.0599	0.1433	97.96%	94.29%
19	0.0811	0.1256	97.34%	94.29%
20	0.0620	0.1277	98.57%	94.29%

QRNN Clean

Metric	Value
Accuracy	65.71%
Precision	68.53%
Recall	65.71%
F1 Score	66.48%
Loss	0.6643
Train Loss	0.5414
Val Loss	0.6643
Train Accuracy	88.75%
Val Accuracy	78.57%
Train Size	489
Val Size	70
Test Size	140

QRNN Clean Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	61	31
Class 1	17	31

QRNN Clean Classification Report

Class	Precision	Recall	F1 Score	Support
0	78.21%	66.30%	71.76%	92.0000
1	50.00%	64.58%	56.36%	48.0000
macro avg	64.10%	65.44%	64.06%	140.0000
weighted avg	68.53%	65.71%	66.48%	140.0000
Accuracy			65.71%	

QRNN Clean Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.6948	0.6979	67.08%	67.14%
2	0.6633	0.6839	73.21%	64.29%
3	0.6379	0.6687	78.94%	74.29%
4	0.6207	0.6612	82.62%	74.29%
5	0.6046	0.6585	85.48%	75.71%
6	0.5952	0.6546	86.09%	75.71%
7	0.5849	0.6536	87.32%	75.71%
8	0.5765	0.6517	87.32%	75.71%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
9	0.5699	0.6522	87.93%	75.71%
10	0.5655	0.6516	87.53%	75.71%
11	0.5595	0.6529	87.93%	75.71%
12	0.5582	0.6530	87.93%	77.14%
13	0.5514	0.6547	88.14%	77.14%
14	0.5506	0.6555	88.14%	78.57%
15	0.5469	0.6583	88.14%	77.14%
16	0.5449	0.6599	87.93%	78.57%
17	0.5432	0.6609	88.34%	78.57%
18	0.5429	0.6619	88.55%	78.57%
19	0.5432	0.6623	88.34%	78.57%
20	0.5414	0.6643	88.75%	78.57%